Geant4-Agent Full Test Report with Fuzzy-Language Robustness

Automated Evaluation Pipeline

2026-03-05

1 Run Config

- config: `nlu/bert_lab/configs/deepseek_api.local.json`
- provider: `deepseek`
- base_url: `https://api.deepseek.com`
- model: `deepseek-chat`
- lang: `zh`, min_confidence=0.6

2 Summary Metrics

metric	value
total_scenarios	17.0000
strict_pass_rate	0.8235
avg_accuracy	0.9069
avg_naturalness	1.0000
avg_grounding_mismatch_rate	0.0000
closure_rate	0.8824
avg_turns_per_scenario	1.6471
clarification_turn_ratio	0.3929
overwrite_guard_pass_rate	1.0000
contradiction_detect_rate	1.0000
fuzzy_pass_rate	0.7778
fuzzy_closure_rate	0.7778
failed_ids	S08, F01, F07

3 Scenario Table

id	category	domain	turns	accuracy	closure	pass
S01	full_closure	Geometry+Source+Physics+Output	1	1.000	True	True
S02	geometry_family	Geometry family	2	1.000	True	True
S03	overwrite_control	Overwrite safety	3	1.000	True	True
S04	question_safety	Question turn behavior	2	1.000	True	True
S05	source_mode	Source mode	1	1.000	True	True
S06	source_mode	Source mode	1	1.000	True	True
S07	consultative	Consultative flow	2	1.000	True	True
S08	output_update	Output update	2	0.667	True	False
F01	fuzzy_language	Fuzzy language	2	0.750	True	False
F02	fuzzy_language	Fuzzy language	1	1.000	True	True
F03	fuzzy_contradiction	Contradiction handling	2	1.000	True	True
F04	fuzzy_question	Question safety	2	1.000	True	True
F05	fuzzy_incomplete	Novice request	1	1.000	False	True
F06	fuzzy_units	Unit ambiguity	1	1.000	True	True
F07	fuzzy_output	Output update	2	0.000	True	False
F08	fuzzy_consultative	Consultative flow	2	1.000	True	True
F09	fuzzy_multimodal_stub	Pseudo multimodal text	1	1.000	False	True

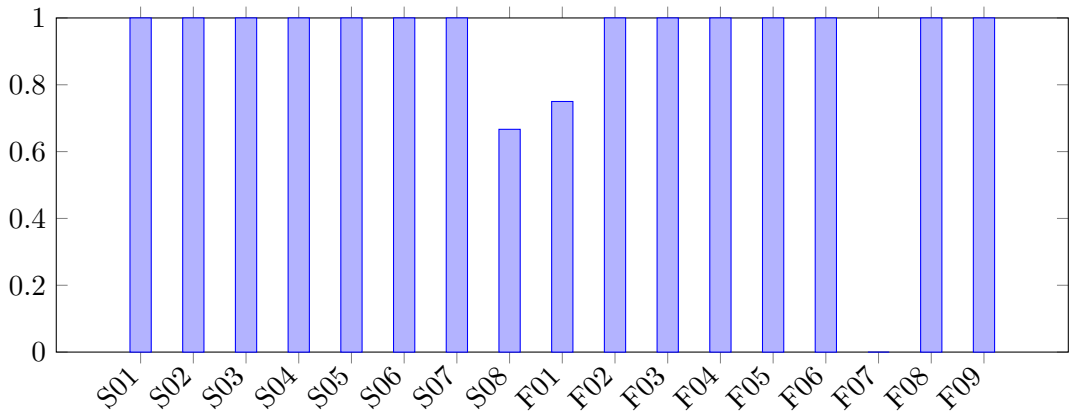
4 Category Table

legend	category	n	pass_rate	closure_rate	avg_acc	avg_nat
C01	consultative	1	1.000	1.000	1.000	1.000
C02	full_closure	1	1.000	1.000	1.000	1.000
C03	fuzzy_consultative	1	1.000	1.000	1.000	1.000
C04	fuzzy_contradiction	1	1.000	1.000	1.000	1.000
C05	fuzzy_incomplete	1	1.000	0.000	1.000	1.000
C06	fuzzy_language	2	0.500	1.000	0.875	1.000
C07	fuzzy_multimodal_stub	1	1.000	0.000	1.000	1.000
C08	fuzzy_output	1	0.000	1.000	0.000	1.000
C09	fuzzy_question	1	1.000	1.000	1.000	1.000
C10	fuzzy_units	1	1.000	1.000	1.000	1.000
C11	geometry_family	1	1.000	1.000	1.000	1.000
C12	output_update	1	0.000	1.000	0.667	1.000

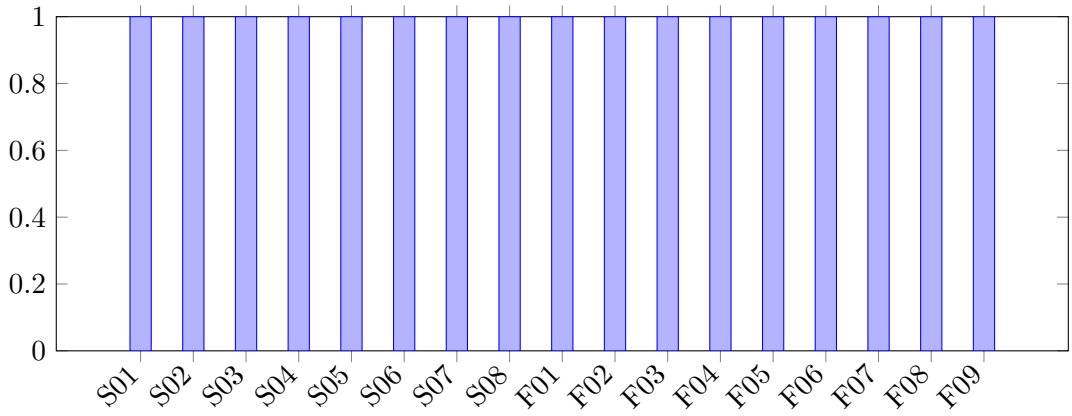
C13	overwrite_control	1	1.000	1.000	1.000	1.000
C14	question_safety	1	1.000	1.000	1.000	1.000
C15	source_mode	2	1.000	1.000	1.000	1.000

5 Visual Analysis

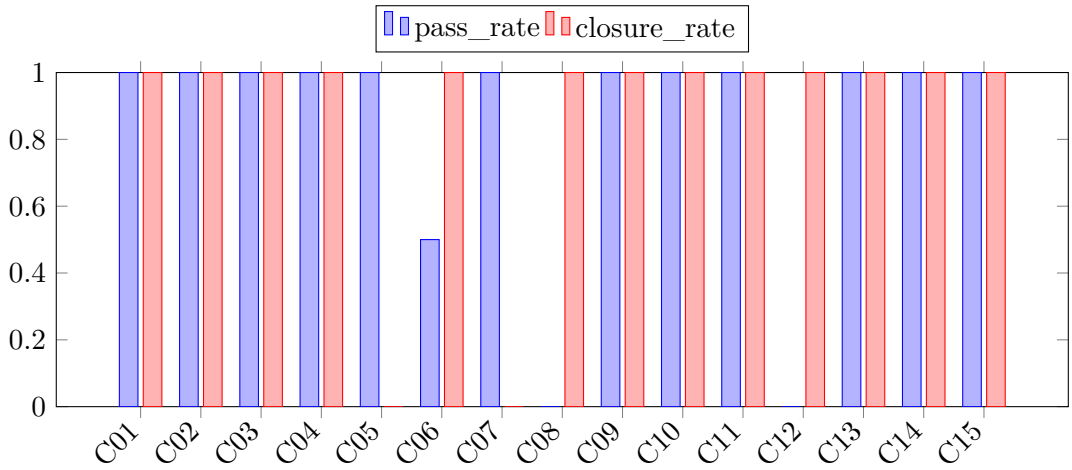
5.1 Scenario Accuracy



5.2 Scenario Naturalness



5.3 Category Pass vs Closure



6 Representative Raw Dialogues (1 per category)

6.1 consultative / S07: Consult then commit

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "This is pure gamma attenuation in copper, no hadrons. Recommend a
        Geant4 physics list and explain briefly."
  },
  {
    "role": "assistant",
    "content": "当前已提交配置为：物理列表 FTFP_BERT，输出格式 未指定，源类型 未指定，粒
        子 gamma。如需修改，请直接说明要改哪一项。"
  },
  {
    "role": "user",
    "content": "Adopt your recommendation. Geometry is 1m copper cube, gamma point 1
        MeV at center along +z, output json."
  },
  {
    "role": "assistant",
    "content": "好的，配置已经全部完成。根据您的设定，我们创建了一个边长为1米的铜立方体几
        何结构，并在其中心放置了一个沿+Z方向发射的1MeV伽马点源，输出格式为JSON。所有参数均
        已确认。"
  }
]
```


Internal iterative trace

```
[
 {
   "turn": 1,
   "user": "This is pure gamma attenuation in copper, no hadrons. Recommend a Geant4
       physics list and explain briefly.",
   "assistant": "当前已提交配置为：物理列表 FTFP_BERT，输出格式 未指定，源类型 未指定，
       粒子 gamma。如需修改，请直接说明要改哪一项。",
   "dialogue_action": "summarize_progress",
   "asked_fields_friendly": [
     "几何结构类型",
     "几何参数 module_x"
   ],
   "missing_fields": [
     "geometry.structure",
     "geometry.params.module_x",
     "geometry.params.module_y",
     "geometry.params.module_z",
     "materials.volume_material_map",
     "source.type",
     "source.energy",
     "source.position",
     "source.direction",
     "output.format",
     "output.path"
   ],
   "normalization": {
     "intent": "SET",
     "confidence": 1.0,
     "backend": "llm_slot_frame"
   },
   "llm_used": true,
   "fallback_reason": null,
   "llm_stage_failures": [],
   "rejected_updates": [],
   "applied_rules": [
     {
       "path": "materials.selected_materials",
       "rule": "winner_selected",
       "producer": "slot_mapper",
       "confidence": 1.0
     },
     {
       "path": "source.particle",
       "rule": "winner_selected",
       "producer": "slot_mapper",
```

```
      "confidence": 1.0
    },
    {
      "path": "physics.physics_list",
      "rule": "winner_selected",
      "producer": "llm_recommender",
      "confidence": 0.95
    },
    {
      "path": "physics.backup_physics_list",
      "rule": "winner_selected",
      "producer": "llm_recommender",
      "confidence": 0.95
    },
    {
      "path": "physics.selection_reasons",
      "rule": "winner_selected",
      "producer": "llm_recommender",
      "confidence": 0.95
    },
    {
      "path": "physics.covered_processes",
      "rule": "winner_selected",
      "producer": "llm_recommender",
      "confidence": 0.95
    },
    {
      "path": "physics.selection_source",
      "rule": "winner_selected",
      "producer": "llm_recommender",
      "confidence": 0.95
    },
    {
      "rule": "post_commit_semantic_sync",
      "count": 4
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Adopt your recommendation. Geometry is 1m copper cube, gamma point 1MeV
      at center along +z, output json.",
  "assistant": "好的，配置已经全部完成。根据您的设定，我们创建了一个边长为1米的铜立方体
      几何结构，并在其中心放置了一个沿+Z方向发射的1MeV伽马点源，输出格式为JSON。所有参数
```

```
    均已确认。",
"dialogue_action": "finalize",
"asked_fields_friendly": [],
"missing_fields": [],
"normalization": {
  "intent": "SET",
  "confidence": 1.0,
  "backend": "llm_slot_frame"
},
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [
  {
    "path": "materials.selected_materials",
    "producer": "slot_mapper",
    "reason_code": "E_LOCKED_FIELD_OVERRIDE",
    "detail": "attempt to modify locked field"
  },
  {
    "path": "source.particle",
    "producer": "slot_mapper",
    "reason_code": "E_LOCKED_FIELD_OVERRIDE",
    "detail": "attempt to modify locked field"
  },
  {
    "path": "output.format",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  }
],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_x",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
```

```
      {
        "path": "geometry.params.module_y",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "geometry.params.module_z",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "source.type",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "source.energy",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "source.position",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "source.direction",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      }
    ],
    "grounding_mismatch": false
  }
]
```


Checks


```
[
  {
```

```
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "physics.physics_list",
    "expected": "FTFP_BERT",
    "actual": "FTFP_BERT",
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "json",
    "actual": "json",
    "ok": true
  }
]
```


6.2 full_closure / S01: Single-turn full closure

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Need a 1m x 1m x 1m copper box, gamma point source 1 MeV at center
        along +z, physics FTFP_BERT, output json."
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Need a 1m x 1m x 1m copper box, gamma point source 1 MeV at center along
        +z, physics FTFP_BERT, output json.",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
```

```
"normalization": {
  "intent": "SET",
  "confidence": 1.0,
  "backend": "llm_slot_frame"
},
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [
  {
    "path": "physics.physics_list",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  },
  {
    "path": "output.format",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  }
],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_x",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_y",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_z",
    "rule": "winner_selected",
    "producer": "slot_mapper",
```

```
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
}
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "geometry.structure",
    "expected": "single_box",
    "actual": "single_box",
    "ok": true
```

```
  },
  {
    "field": "source.type",
    "expected": "point",
    "actual": "point",
    "ok": true
  },
  {
    "field": "source.particle",
    "expected": "gamma",
    "actual": "gamma",
    "ok": true
  },
  {
    "field": "physics.physics_list",
    "expected": "FTFP_BERT",
    "actual": "FTFP_BERT",
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "json",
    "actual": "json",
    "ok": true
  }
]
```


6.3 fuzzy_consultative / F08: Consultative with fuzzy wording

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "I only care about gamma attenuation in copper and no hadrons.
        Recommend a safe Geant4 physics list."
  },
  {
    "role": "assistant",
    "content": "好的，当前配置进度如下：\n\n**已完成：**\n* **材料**：已选定铜。\n* **源
        **：已确定粒子类型为伽马射线。\n* **物理**：已根据您的要求（仅关心伽马在铜中的衰
        减，无需强子物理），推荐并设置了安全的物理列表。\n\n**待补充：**\n* **几何**：需要
        您提供探测器的几何结构类型和具体尺寸。\n* **材料绑定**：需要明确将铜材料具体绑定到
        哪个几何体积上。\n* **源细节**：需要设定源的类型（如点源、面源等）和能量。\n* **输
        出**：需要指定输出数据的格式和保存路径。\n\n请先补充几何结构的相关信息。"
  },
```

```
  {
    "role": "user",
    "content": "Use your recommendation. Geometry 1m copper cube, gamma point 1MeV at
        center +z, output json."
  },
  {
    "role": "assistant",
    "content": "好的，配置已经全部完成！\n\n根据您最新的要求，我已为您整合并确认了以下设
        置：\n* **几何**：一个边长为1米的铜立方体。\n* **源**：中心位置、沿+Z方向发射的
        1MeV伽马点源。\n* **物理**：采用推荐的、安全的物理列表（仅包含伽马相关过程）。\n*
        **输出**：结果将保存为JSON格式。\n\n所有参数均已就绪，可以开始运行模拟了。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "I only care about gamma attenuation in copper and no hadrons. Recommend a
         safe Geant4 physics list.",
    "assistant": "好的，当前配置进度如下：\n\n**已完成：**\n* **材料**：已选定铜。\n* **
        源**：已确定粒子类型为伽马射线。\n* **物理**：已根据您的要求（仅关心伽马在铜中的衰
        减，无需强子物理），推荐并设置了安全的物理列表。\n\n**待补充：**\n* **几何**：需要
        您提供探测器的几何结构类型和具体尺寸。\n* **材料绑定**：需要明确将铜材料具体绑定到
        哪个几何体积上。\n* **源细节**：需要设定源的类型（如点源、面源等）和能量。\n* **输
        出**：需要指定输出数据的格式和保存路径。\n\n请先补充几何结构的相关信息。",
    "dialogue_action": "summarize_progress",
    "asked_fields_friendly": [
      "几何结构类型",
      "几何参数 module_x"
    ],
    "missing_fields": [
      "geometry.structure",
      "geometry.params.module_x",
      "geometry.params.module_y",
      "geometry.params.module_z",
      "materials.volume_material_map",
      "source.type",
      "source.energy",
      "source.position",
      "source.direction",
      "output.format",
      "output.path"
    ],
```

```
"normalization": {
  "intent": "MODIFY",
  "confidence": 0.95,
  "backend": "llm_slot_frame"
},
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [],
"applied_rules": [
  {
    "path": "materials.selected_materials",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 0.95
  },
  {
    "path": "source.particle",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 0.95
  },
  {
    "path": "physics.physics_list",
    "rule": "winner_selected",
    "producer": "llm_recommender",
    "confidence": 0.95
  },
  {
    "path": "physics.backup_physics_list",
    "rule": "winner_selected",
    "producer": "llm_recommender",
    "confidence": 0.95
  },
  {
    "path": "physics.selection_reasons",
    "rule": "winner_selected",
    "producer": "llm_recommender",
    "confidence": 0.95
  },
  {
    "path": "physics.covered_processes",
    "rule": "winner_selected",
    "producer": "llm_recommender",
    "confidence": 0.95
```

```
    },
    {
      "path": "physics.selection_source",
      "rule": "winner_selected",
      "producer": "llm_recommender",
      "confidence": 0.95
    },
    {
      "rule": "post_commit_semantic_sync",
      "count": 4
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Use your recommendation. Geometry 1m copper cube, gamma point 1MeV at
      center +z, output json.",
  "assistant": "好的，配置已经全部完成！\n\n根据您最新的要求，我已为您整合并确认了以下设
      置：\n* **几何**：一个边长为1米的铜立方体。\n* **源**：中心位置、沿+Z方向发射的
      1MeV伽马点源。\n* **物理**：采用推荐的、安全的物理列表（仅包含伽马相关过程）。\n*
      **输出**：结果将保存为JSON格式。\n\n所有参数均已就绪，可以开始运行模拟了。",
  "dialogue_action": "finalize",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "SET",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [
    {
      "path": "materials.selected_materials",
      "producer": "slot_mapper",
      "reason_code": "E_LOCKED_FIELD_OVERRIDE",
      "detail": "attempt to modify locked field"
    },
    {
      "path": "source.particle",
      "producer": "slot_mapper",
      "reason_code": "E_LOCKED_FIELD_OVERRIDE",
      "detail": "attempt to modify locked field"
```

```
    },
    {
      "path": "output.format",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    }
  ],
  "applied_rules": [
    {
      "path": "geometry.structure",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_x",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
```

```
      {
        "path": "source.position",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "source.direction",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      }
    ],
    "grounding_mismatch": false
  }
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "json",
    "actual": "json",
    "ok": true
  }
]
```


6.4 fuzzy_contradiction / F03: Contradictory rewrite in one turn

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json."
  },
  {
    "role": "assistant",
```

```
    "content": "好的，你的 Geant4 配置已经完成了！\n\n根据你提供的信息，我们已经设定了：\
        n* **几何**：一个 1m x 1m x 1m 的立方体模块。\n* **材料**：立方体材料为铜 (Copper
        )。\n* **物理过程**：使用 FTFP_BERT 物理列表。\n* **粒子源**：在中心位置，沿 +Z
        方向发射 1 MeV 能量的伽马射线。\n* **输出**：结果将保存为 JSON 格式。\n\n所有必要
        参数都已确认，可以准备运行模拟了。"
  },
  {
    "role": "user",
    "content": "Change geometry to cylinder radius 30mm, but keep the 1m cube
        unchanged."
  },
  {
    "role": "assistant",
    "content": "检测到要修改已确认的几何结构。请确认是否将几何结构类型从“立方体”改为“圆
        柱体”？"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json.",
    "assistant": "好的，你的 Geant4 配置已经完成了！\n\n根据你提供的信息，我们已经设定
        了：\n* **几何**：一个 1m x 1m x 1m 的立方体模块。\n* **材料**：立方体材料为铜 (
        Copper)。\n* **物理过程**：使用 FTFP_BERT 物理列表。\n* **粒子源**：在中心位置，
        沿 +Z 方向发射 1 MeV 能量的伽马射线。\n* **输出**：结果将保存为 JSON 格式。\n\n所
        有必要参数都已确认，可以准备运行模拟了。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
    "rejected_updates": [
      {
        "path": "physics.physics_list",
        "producer": "slot_mapper",
```

```
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    },
    {
      "path": "output.format",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    }
  ],
  "applied_rules": [
    {
      "path": "geometry.structure",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_x",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
```

```
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Change geometry to cylinder radius 30mm, but keep the 1m cube unchanged
      .",
  "assistant": "检测到要修改已确认的几何结构。请确认是否将几何结构类型从“立方体”改为
      “圆柱体”？",
  "dialogue_action": "confirm_overwrite",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "MODIFY",
    "confidence": 0.95,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [
    {
      "path": "geometry.params.module_x",
      "producer": "slot_mapper",
      "reason_code": "E_LOCKED_FIELD_OVERRIDE",
      "detail": "attempt to modify locked field"
    },
    {
      "path": "geometry.params.module_y",
      "producer": "slot_mapper",
      "reason_code": "E_LOCKED_FIELD_OVERRIDE",
```

```
      "detail": "attempt to modify locked field"
    },
    {
      "path": "geometry.params.module_z",
      "producer": "slot_mapper",
      "reason_code": "E_LOCKED_FIELD_OVERRIDE",
      "detail": "attempt to modify locked field"
    }
  ],
  "applied_rules": [
    {
      "rule": "pending_overwrite_confirmation_required",
      "count": 1
    },
    {
      "path": "geometry.params.child_rmax",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 0.95
    }
  ],
  "grounding_mismatch": false
}
]
```


Checks


```
[
  {
    "field": "geometry.structure",
    "expected": "single_box",
    "actual": "single_box",
    "ok": true
  },
  {
    "field": "has_any_action",
    "expected": [
      "confirm_overwrite",
      "ask_clarification"
    ],
    "actual": [
      "confirm_overwrite"
    ],
    "ok": true
  }
```

]

6.5 fuzzy_incomplete / F05: Incomplete novice input

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Can you set up a copper gamma test and give me a runnable baseline?"
  },
  {
    "role": "assistant",
    "content": "目前配置进度如下：\n\n**几何**：待补充结构类型和尺寸参数。 \n**材料**：已
        选定铜，并记录了选择依据和来源；待补充材料与体积的绑定关系。 \n**源**：已确定使用
        伽马射线，并记录了选择依据和来源；待补充具体的源类型和能量。 \n**物理**：待选择物
        理列表。 \n**输出**：待指定输出格式和保存路径。\n\n请继续补充待完善的信息，以便生
        成可运行的基线配置。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Can you set up a copper gamma test and give me a runnable baseline?",
    "assistant": "目前配置进度如下：\n\n**几何**：待补充结构类型和尺寸参数。 \n**材料**：
        已选定铜，并记录了选择依据和来源；待补充材料与体积的绑定关系。 \n**源**：已确定使
        用伽马射线，并记录了选择依据和来源；待补充具体的源类型和能量。 \n**物理**：待选择
        物理列表。 \n**输出**：待指定输出格式和保存路径。\n\n请继续补充待完善的信息，以便
        生成可运行的基线配置。",
    "dialogue_action": "summarize_progress",
    "asked_fields_friendly": [
      "几何结构类型",
      "几何参数 module_x"
    ],
    "missing_fields": [
      "geometry.structure",
      "geometry.params.module_x",
      "geometry.params.module_y",
      "geometry.params.module_z",
      "materials.volume_material_map",
      "source.type",
      "source.energy",
      "source.position",
```

```
      "source.direction",
      "physics.physics_list",
      "output.format",
      "output.path"
    ],
    "normalization": {
      "intent": "QUESTION",
      "confidence": 0.95,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
    "rejected_updates": [],
    "applied_rules": [
      {
        "path": "materials.selected_materials",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 0.95
      },
      {
        "path": "source.particle",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 0.95
      },
      {
        "rule": "post_commit_semantic_sync",
        "count": 4
      }
    ],
    "grounding_mismatch": false
  }
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": false,
    "actual": false,
    "ok": true
  },
```

```
  {
    "field": "missing_at_least",
    "expected": 1,
    "actual": 12,
    "ok": true
  },
  {
    "field": "has_any_action",
    "expected": [
      "ask_clarification",
      "summarize_progress",
      "answer_status"
    ],
    "actual": [
      "summarize_progress"
    ],
    "ok": true
  }
]
```


6.6 fuzzy_language / F02: Code-switch shorthand

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "target: Cu cube 1000mm^3 each side; src gamma point E=1MeV pos(0,0,0)
        dir(0,0,1); physics=FTFP_BERT; output=root"
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "target: Cu cube 1000mm^3 each side; src gamma point E=1MeV pos(0,0,0) dir
        (0,0,1); physics=FTFP_BERT; output=root",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
```

```
"missing_fields": [],
"normalization": {
  "intent": "SET",
  "confidence": 1.0,
  "backend": "llm_slot_frame"
},
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [
  {
    "path": "physics.physics_list",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  },
  {
    "path": "output.format",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  }
],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_x",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_y",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_z",
    "rule": "winner_selected",
```

```
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
}
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "geometry.structure",
    "expected": "single_box",
    "actual": "single_box",
```

```
    "ok": true
  },
  {
    "field": "source.type",
    "expected": "point",
    "actual": "point",
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "root",
    "actual": "root",
    "ok": true
  }
]
```


6.7 fuzzy_multimodal_stub / F09: Multimodal placeholder text

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Follow this sketch: [image: copper cube, center source, +z direction],
         build config first."
  },
  {
    "role": "assistant",
    "content": "当前配置进度如下：\n\n几何部分已设定好结构类型和X、Y方向尺寸；材料部分已选
        定具体材料并完成与体积的绑定；源部分已设定类型、能量和位置，还需补充粒子类型；物理
        列表以及输出格式、路径尚待补充。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Follow this sketch: [image: copper cube, center source, +z direction],
        build config first.",
    "assistant": "当前配置进度如下：\n\n几何部分已设定好结构类型和X、Y方向尺寸；材料部分
        已选定具体材料并完成与体积的绑定；源部分已设定类型、能量和位置，还需补充粒子类型；
        物理列表以及输出格式、路径尚待补充。",
    "dialogue_action": "summarize_progress",
    "asked_fields_friendly": [
```

```
    "粒子类型",
    "物理列表"
  ],
  "missing_fields": [
    "source.particle",
    "physics.physics_list",
    "output.format",
    "output.path"
  ],
  "normalization": {
    "intent": "SET",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [],
  "applied_rules": [
    {
      "path": "geometry.structure",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_x",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
```

```
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.position",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
}
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": false,
    "actual": false,
    "ok": true
  },
  {
    "field": "missing_at_least",
    "expected": 1,
    "actual": 4,
    "ok": true
  },
  {
    "field": "has_any_action",
    "expected": [
```

```
      "ask_clarification",
      "answer_status",
      "summarize_progress"
    ],
    "actual": [
      "summarize_progress"
    ],
    "ok": true
  }
]
```


6.8 fuzzy_output / F07: Colloquial output rewrite

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json."
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  },
  {
    "role": "user",
    "content": "I want the ROOT tree style output file."
  },
  {
    "role": "assistant",
    "content": "检测到您想要修改输出格式。请确认是否将输出格式从 json 改为 root？"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json.",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
```

```
"normalization": {
  "intent": "SET",
  "confidence": 1.0,
  "backend": "llm_slot_frame"
},
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [
  {
    "path": "physics.physics_list",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  },
  {
    "path": "output.format",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  }
],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_x",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_y",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_z",
    "rule": "winner_selected",
    "producer": "slot_mapper",
```

```
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "I want the ROOT tree style output file.",
  "assistant": "检测到您想要修改输出格式。请确认是否将输出格式从 json 改为 root? ",
  "dialogue_action": "confirm_overwrite",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "SET",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [],
```

```
    "applied_rules": [
      {
        "rule": "pending_overwrite_confirmation_required",
        "count": 1
      }
    ],
    "grounding_mismatch": false
  }
]
```


Checks


```
[
  {
    "field": "output.format",
    "expected": "root",
    "actual": "json",
    "ok": false
  }
]
```


6.9 fuzzy_question / F04: Fuzzy question should not overwrite

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Setup 1m copper cube, gamma 1MeV point source at center +z, FTFP_BERT,
        json output."
  },
  {
    "role": "assistant",
    "content": "配置已完成，所有参数已根据你的输入设置完毕。"
  },
  {
    "role": "user",
    "content": "Would proton be better?"
  },
  {
    "role": "assistant",
    "content": "我为你解释一下当前配置的依据：\n\n**物理列表**：根据你明确指定的“
        FTFP_BERT”设置。\n**材料**：根据你明确提供的“1m copper cube”设置。\n**源**：根
        据你明确描述的“gamma 1MeV point source at center +z”设置。\n\n以上所有参数都已
        确认并生效。"
  }
```

]

Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Setup 1m copper cube, gamma 1MeV point source at center +z, FTFP_BERT,
        json output.",
    "assistant": "配置已完成，所有参数已根据你的输入设置完毕。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
    "rejected_updates": [
      {
        "path": "physics.physics_list",
        "producer": "slot_mapper",
        "reason_code": "E_LOWER_PRIORITY",
        "detail": "superseded by winner"
      },
      {
        "path": "output.format",
        "producer": "slot_mapper",
        "reason_code": "E_LOWER_PRIORITY",
        "detail": "superseded by winner"
      }
    ],
    "applied_rules": [
      {
        "path": "geometry.structure",
        "rule": "winner_selected",
        "producer": "slot_mapper",
        "confidence": 1.0
      },
      {
        "path": "geometry.params.module_x",
        "rule": "winner_selected",
```

```
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Would proton be better?",
```

```
    "assistant": "我为你解释一下当前配置的依据：\n\n**物理列表**：根据你明确指定的“
        FTFP_BERT”设置。\n**材料**：根据你明确提供的“1m copper cube”设置。\n**源**：根
        据你明确描述的“gamma 1MeV point source at center +z”设置。\n\n以上所有参数都已
        确认并生效。",
    "dialogue_action": "explain_choice",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "QUESTION",
      "confidence": 0.9,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
    "rejected_updates": [
      {
        "path": "source.particle",
        "producer": "slot_mapper",
        "reason_code": "E_OVERWRITE_WITHOUT_EXPLICIT_USER_INTENT",
        "detail": "overwrite blocked: user did not explicitly request this field update
            "
      }
    ],
    "applied_rules": [],
    "grounding_mismatch": false
  }
]
```


Checks


```
[
  {
    "field": "source.particle",
    "expected": "gamma",
    "actual": "gamma",
    "ok": true
  },
  {
    "field": "has_any_action",
    "expected": [
      "explain_choice",
      "answer_status",
      "summarize_progress"
    ],
```

```
    "actual": [
      "explain_choice"
    ],
    "ok": true
  }
]
```


6.10 fuzzy_units / F06: Mixed units in one sentence

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Copper target 100cm x 1000mm x 1m, gamma point source 1000keV at
        center along +z, physics FTFP_BERT, output json."
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Copper target 100cm x 1000mm x 1m, gamma point source 1000keV at center
        along +z, physics FTFP_BERT, output json.",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
    "rejected_updates": [
      {
        "path": "physics.physics_list",
        "producer": "slot_mapper",
```

```
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    },
    {
      "path": "output.format",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    }
  ],
  "applied_rules": [
    {
      "path": "geometry.structure",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_x",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
```

```
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
}
]
```


Checks


```
[
  {
    "field": "source.particle",
    "expected": "gamma",
    "actual": "gamma",
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "json",
    "actual": "json",
    "ok": true
  }
]
```


6.11 geometry_family / S02: Two-turn tubs setup

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Need a copper cylinder, radius 30 mm, half-length 50 mm, gamma point
        source 2 MeV at (0,0,-100) toward (0,0,1)."
  },
```

```
  {
    "role": "assistant",
    "content": "当前配置进度如下：\n\n几何部分：已根据你的输入，更新了结构类型和圆柱体的半
        径、半高参数。 \n材料部分：已选定铜材料，并完成了与体积的绑定。 \n源部分：已设置为
        伽马点源，能量2 MeV，位置和方向已同步。 \n\n接下来还需要补充以下设置才能完成配置：
         \n- 物理列表 \n- 输出格式 \n- 输出路径 \n\n请提供这些信息，我们可以继续推进。"
  },
  {
    "role": "user",
    "content": "Use QBBC, output json."
  },
  {
    "role": "assistant",
    "content": "好的，配置已完成。\n\n根据你的设置，物理列表已选定为 QBBC，输出格式为
        JSON，输出路径也已确认。所有必要信息都已齐备，可以开始生成配置文件了。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Need a copper cylinder, radius 30 mm, half-length 50 mm, gamma point
        source 2 MeV at (0,0,-100) toward (0,0,1).",
    "assistant": "当前配置进度如下：\n\n几何部分：已根据你的输入，更新了结构类型和圆柱体的
        半径、半高参数。 \n材料部分：已选定铜材料，并完成了与体积的绑定。 \n源部分：已设置
        为伽马点源，能量2 MeV，位置和方向已同步。 \n\n接下来还需要补充以下设置才能完成配
        置： \n- 物理列表 \n- 输出格式 \n- 输出路径 \n\n请提供这些信息，我们可以继续推
        进。",
    "dialogue_action": "summarize_progress",
    "asked_fields_friendly": [
      "物理列表",
      "输出格式"
    ],
    "missing_fields": [
      "physics.physics_list",
      "output.format",
      "output.path"
    ],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
```

```
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.child_rmax",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.child_hz",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "materials.selected_materials",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "source.type",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "source.particle",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "source.energy",
    "rule": "winner_selected",
    "producer": "slot_mapper",
```

```
      "confidence": 1.0
    },
    {
      "path": "source.position",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Use QBBC, output json.",
  "assistant": "好的，配置已完成。\n\n根据你的设置，物理列表已选定为 QBBC，输出格式为
      JSON，输出路径也已确认。所有必要信息都已齐备，可以开始生成配置文件了。",
  "dialogue_action": "finalize",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "SET",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [
    {
      "path": "physics.physics_list",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    },
    {
      "path": "output.format",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    }
  ],
  "applied_rules": [
    {
      "path": "physics.physics_list",
      "rule": "winner_selected",
```

```
      "producer": "user_explicit",
      "confidence": 1.0
    },
    {
      "path": "output.format",
      "rule": "winner_selected",
      "producer": "user_explicit",
      "confidence": 1.0
    },
    {
      "rule": "post_commit_semantic_sync",
      "count": 3
    }
  ],
  "grounding_mismatch": false
}
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "geometry.structure",
    "expected": "single_tubs",
    "actual": "single_tubs",
    "ok": true
  },
  {
    "field": "physics.physics_list",
    "expected": "QBBC",
    "actual": "QBBC",
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "json",
    "actual": "json",
    "ok": true
  }
```

]

6.12 output_update / S08: Output-only change

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json."
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  },
  {
    "role": "user",
    "content": "Change output to root."
  },
  {
    "role": "assistant",
    "content": "检测到您要将输出格式从 json 改为 root，这会覆盖已确认的设置。请确认是否应
        用此修改？"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json.",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
```

```
"rejected_updates": [
  {
    "path": "physics.physics_list",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  },
  {
    "path": "output.format",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  }
],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_x",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_y",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_z",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "materials.selected_materials",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
```

```
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Change output to root.",
  "assistant": "检测到您要将输出格式从 json 改为 root，这会覆盖已确认的设置。请确认是否
      应用此修改？",
  "dialogue_action": "confirm_overwrite",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "MODIFY",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [],
  "applied_rules": [
    {
      "rule": "pending_overwrite_confirmation_required",
      "count": 1
    }
  ],
  "grounding_mismatch": false
```

```
  }
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "root",
    "actual": "json",
    "ok": false
  },
  {
    "field": "source.particle",
    "expected": "gamma",
    "actual": "gamma",
    "ok": true
  }
]
```


6.13 overwrite_control / S03: Confirm overwrite flow

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json."
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  },
  {
    "role": "user",
    "content": "Change material to G4_Al."
  },
  {
    "role": "assistant",
```

```
    "content": "检测到需要覆盖已确认的材料设置。请确认是否应用以下修改：将材料从 G4_Cu 改
        为 G4_Al。"
  },
  {
    "role": "user",
    "content": "Confirm."
  },
  {
    "role": "assistant",
    "content": "配置已更新完成。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "Copper cube 1m x 1m x 1m, gamma point 1 MeV at center +z, physics
        FTFP_BERT, output json.",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [],
    "rejected_updates": [
      {
        "path": "physics.physics_list",
        "producer": "slot_mapper",
        "reason_code": "E_LOWER_PRIORITY",
        "detail": "superseded by winner"
      },
      {
        "path": "output.format",
        "producer": "slot_mapper",
        "reason_code": "E_LOWER_PRIORITY",
        "detail": "superseded by winner"
      }
```

```
  ],
  "applied_rules": [
    {
      "path": "geometry.structure",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_x",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
```

```
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "Change material to G4_Al.",
  "assistant": "检测到需要覆盖已确认的材料设置。请确认是否应用以下修改：将材料从 G4_Cu
      改为 G4_Al。",
  "dialogue_action": "confirm_overwrite",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "MODIFY",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [],
  "applied_rules": [
    {
      "rule": "pending_overwrite_confirmation_required",
      "count": 1
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 3,
  "user": "Confirm.",
  "assistant": "配置已更新完成。",
  "dialogue_action": "finalize",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "CONFIRM",
    "confidence": 1.0,
    "backend": "pending_overwrite_confirmation"
  },
```

```
    "llm_used": true,
    "fallback_reason": null,
    "llm_stage_failures": [
      "E_LLM_SLOT_EMPTY"
    ],
    "rejected_updates": [],
    "applied_rules": [
      {
        "path": "materials.selected_materials",
        "rule": "winner_selected",
        "producer": "user_explicit",
        "confidence": 1.0
      },
      {
        "rule": "post_commit_semantic_sync",
        "count": 1
      }
    ],
    "grounding_mismatch": false
  }
]
```


Checks


```
[
  {
    "field": "complete",
    "expected": true,
    "actual": true,
    "ok": true
  },
  {
    "field": "materials.selected_materials",
    "expected": [
      "G4_Al"
    ],
    "actual": [
      "G4_Al"
    ],
    "ok": true
  },
  {
    "field": "has_action",
    "expected": "confirm_overwrite",
    "actual": true,
```

```
    "ok": true
  }
]
```


6.14 question_safety / S04: Question should not overwrite

User-facing raw dialogue


```
[
  {
    "role": "user",
    "content": "1m copper box, gamma point source 1 MeV at origin along +z, physics
        FTFP_BERT, output json."
  },
  {
    "role": "assistant",
    "content": "配置已完成。"
  },
  {
    "role": "user",
    "content": "what if proton?"
  },
  {
    "role": "assistant",
    "content": "以下是您当前配置中各项选择的说明：\n\n**物理列表**：根据您明确指定的参数
        选定。 \n**材料**：根据您明确提供的材料信息选定。 \n**源类型**：根据您明确给出的源
        参数选定。"
  }
]
```


Internal iterative trace


```
[
  {
    "turn": 1,
    "user": "1m copper box, gamma point source 1 MeV at origin along +z, physics
        FTFP_BERT, output json.",
    "assistant": "配置已完成。",
    "dialogue_action": "finalize",
    "asked_fields_friendly": [],
    "missing_fields": [],
    "normalization": {
      "intent": "SET",
      "confidence": 1.0,
      "backend": "llm_slot_frame"
    },
```

```
"llm_used": true,
"fallback_reason": null,
"llm_stage_failures": [],
"rejected_updates": [
  {
    "path": "physics.physics_list",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  },
  {
    "path": "output.format",
    "producer": "slot_mapper",
    "reason_code": "E_LOWER_PRIORITY",
    "detail": "superseded by winner"
  }
],
"applied_rules": [
  {
    "path": "geometry.structure",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_x",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_y",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "geometry.params.module_z",
    "rule": "winner_selected",
    "producer": "slot_mapper",
    "confidence": 1.0
  },
  {
    "path": "materials.selected_materials",
    "rule": "winner_selected",
```

```
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
},
{
  "turn": 2,
  "user": "what if proton?",
  "assistant": "以下是您当前配置中各项选择的说明：\n\n**物理列表**：根据您明确指定的参
      数选定。 \n**材料**：根据您明确提供的材料信息选定。 \n**源类型**：根据您明确给出的
      源参数选定。",
  "dialogue_action": "explain_choice",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "QUESTION",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [
    {
      "path": "source.particle",
      "producer": "slot_mapper",
```

```
      "reason_code": "E_OVERWRITE_WITHOUT_EXPLICIT_USER_INTENT",
      "detail": "overwrite blocked: user did not explicitly request this field update
          "
    }
  ],
  "applied_rules": [],
  "grounding_mismatch": false
 }
]
```


Checks


```
[
 {
  "field": "source.particle",
  "expected": "gamma",
  "actual": "gamma",
  "ok": true
 },
 {
  "field": "complete",
  "expected": true,
  "actual": true,
  "ok": true
 }
]
```


6.15 source_mode / S05: Beam source setup

User-facing raw dialogue


```
[
 {
  "role": "user",
  "content": "Silicon cube 50 cm each side, beam gamma source 3 MeV at (0,0,-200)
      direction (0,0,1), physics QBBC, output root."
 },
 {
  "role": "assistant",
  "content": "配置已完成。"
 }
]
```


Internal iterative trace


```
[
```

```
{
  "turn": 1,
  "user": "Silicon cube 50 cm each side, beam gamma source 3 MeV at (0,0,-200)
      direction (0,0,1), physics QBBC, output root.",
  "assistant": "配置已完成。",
  "dialogue_action": "finalize",
  "asked_fields_friendly": [],
  "missing_fields": [],
  "normalization": {
    "intent": "SET",
    "confidence": 1.0,
    "backend": "llm_slot_frame"
  },
  "llm_used": true,
  "fallback_reason": null,
  "llm_stage_failures": [],
  "rejected_updates": [
    {
      "path": "physics.physics_list",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    },
    {
      "path": "output.format",
      "producer": "slot_mapper",
      "reason_code": "E_LOWER_PRIORITY",
      "detail": "superseded by winner"
    }
  ],
  "applied_rules": [
    {
      "path": "geometry.structure",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_x",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_y",
```

```
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "geometry.params.module_z",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "materials.selected_materials",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.type",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.particle",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    },
    {
      "path": "source.energy",
      "rule": "winner_selected",
      "producer": "slot_mapper",
      "confidence": 1.0
    }
  ],
  "grounding_mismatch": false
 }
]
```


Checks


```
[
 {
  "field": "complete",
  "expected": true,
```

```
    "actual": true,
    "ok": true
  },
  {
    "field": "source.type",
    "expected": "beam",
    "actual": "beam",
    "ok": true
  },
  {
    "field": "output.format",
    "expected": "root",
    "actual": "root",
    "ok": true
  }
]
```